

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understandin g of Data (2.5)	Analysis (2.5)	Applicatio n of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightag e	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I S.Vyshnavi(192472041)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Vyshnavi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Q1. Hospital Patient Segmentation

The given SSE values are:

K	SSE
1	600
2	350
3	220
4	200
5	190

In **K-Means clustering**, SSE measures how closely the data points are grouped around their cluster centers. A lower SSE means that the clusters are more compact. However, increasing the number of clusters will always reduce SSE, so we use the **Elbow Method** to select the appropriate value of K.

The SSE decreases significantly from **600 at K=1 to 350 at K=2**, and then to **220 at K=3**. After K=3, the improvement becomes much smaller: SSE decreases only from **220 to 200** and then from **200 to 190**.

Therefore, the **elbow point occurs at K=3**.

Optimal Number of Clusters

The optimal number of clusters is:

K = 3

The three clusters can represent different types of patients based on their medical expenses and health-risk scores.

- **High-risk patients:** Patients having relatively high health-risk scores. These patients may require frequent monitoring and preventive treatment.
- **Moderate-risk patients:** Patients with medium risk who may require regular medical check-ups.
- **Lower-risk/high-expense patients:** Patients who may have higher medical expenses but comparatively lower health-risk scores. Their expenses may need financial or treatment analysis.

Effect on Healthcare Decision-Making

Selecting three clusters helps the healthcare organization understand different patient groups instead of treating all patients in the same way. High-risk patients can receive greater attention and priority. Moderate-risk patients can be monitored through regular healthcare programs, while lower-risk patients can be managed through normal healthcare services.

The clusters can also help hospitals with **resource allocation, patient monitoring, preventive care, treatment planning, and healthcare cost management.**

If K is too small, important patient differences may be hidden. If K is too large, patients may be divided into too many small groups, making the segmentation difficult to manage.

Conclusion

The **optimal number of clusters is 3**, because the SSE decreases substantially up to K=3 and shows only a small improvement after that. Thus, K=3 provides a good balance between accurate patient grouping and simple healthcare management.

Q2. E-Commerce Order Analysis

The dataset contains five orders:

Order	Category	Value
O1	Electronics	₹55,000
O2	Clothing	₹25,000
O3	Electronics	₹57,000
O4	Furniture	₹30,000
O5	Clothing	₹27,000

Hierarchical Agglomerative Clustering (HAC) starts by considering every order as a separate cluster. The algorithm then combines the most similar orders step by step. The resulting combinations are represented using a **dendrogram**.

The two Electronics orders, **O1 and O3**, are very similar because both belong to Electronics and their order values are close: ₹55,000 and ₹57,000. Therefore, they are likely to form a cluster early.

Similarly, O2 and O5 are both Clothing orders and have similar values of ₹25,000 and ₹27,000. Therefore, they are also likely to form a cluster.

O4 belongs to Furniture and has a value of ₹30,000, making it different from the other groups.

Appropriate Number of Clusters

A suitable interpretation is **3 clusters**:

Cluster	Orders	Interpretation
Cluster 1	O1, O3	Electronics, high-value orders
Cluster 2	O2, O5	Clothing, medium-value orders
Cluster 3	O4	Furniture order

The dendrogram can be cut at a level where these three groups are clearly separated.

Significance of the Clusters

The clusters provide useful information about customer purchasing behavior.

Cluster 1 – Electronics:

Customers associated with this group are involved in high-value electronics purchases. The company can provide premium offers, electronic product recommendations, and loyalty benefits.

Cluster 2 – Clothing:

These orders represent customers purchasing clothing products at moderate prices. The retailer can provide clothing discounts, seasonal offers, and personalized recommendations.

Cluster 3 – Furniture:

Furniture purchases have different characteristics and may require separate marketing strategies, delivery planning, and product recommendations.

Business Benefits

The clustering results can help the retailer with:

- Customer segmentation
- Personalized recommendations

- Targeted advertising
- Product promotions
- Inventory planning
- Sales strategy
- Understanding high-value purchases

Conclusion

The appropriate number of clusters is **3**. The clusters separate the orders into meaningful purchasing groups and help the retailer understand different customer purchasing patterns.

Q3. Mobile App User Behavior

The dataset is:

User	Usage Time	Login Frequency	Purchase
U1	120 min	18	₹2,000
U2	60 min	5	₹5,000
U3	140 min	22	₹1,500
U4	50 min	4	₹6,000
U5	130 min	20	₹1,800

The company uses **HAC (Hierarchical Agglomerative Clustering)** to identify groups of users with similar behavior.

In **soft clustering**, users are not forced to belong completely to only one cluster. Instead, each user can have a **membership probability** for different clusters.

For example, the users can be grouped into two broad behavioral patterns.

Cluster 1 – Highly Active Users

U1, U3 and U5 have high daily usage time and high login frequency.

Their membership may be represented as:

User	Active Cluster	High-Spending Cluster
U1	0.90	0.10
U3	0.95	0.05
U5	0.93	0.07

This means U1, U3 and U5 have a strong probability of belonging to the highly active user cluster.

Cluster 2 – High-Spending Users

U2 and U4 have relatively low usage time and login frequency but high in-app purchases.

User	Active Cluster	High-Spending Cluster
U2	0.20	0.80
U4	0.10	0.90

These values indicate that U2 and U4 are more strongly associated with the high-spending group.

Note: The probability values above are illustrative because actual probability-based membership values are not provided in the question.

Soft Clustering vs Hard Clustering

In **hard clustering**, a user belongs to only one cluster.

For example:

U1 → Active User Cluster

In **soft clustering**, the same user can have membership in multiple clusters.

For example:

U1 → 90% Active + 10% High-Spending

Advantages of Soft Clustering

1. **Handles overlapping behavior:** A customer may show characteristics of multiple groups.

2. **More realistic:** Real users do not always fit into one fixed category.
3. **Better personalization:** Different offers can be based on the user's membership probabilities.
4. **Identifies uncertain users:** Users near the boundary between clusters can be detected.
5. **Improves marketing:** Companies can target customers more effectively.
6. **Supports recommendation systems:** Products and services can be recommended according to different behavioral characteristics.

Conclusion

Soft clustering is more flexible than hard clustering because it represents the **degree of membership** of each user. It provides a better understanding of complex customer behavior and can improve personalization, marketing, and user engagement.

Q4. Agricultural Crop Monitoring

The dataset contains:

Farm	Rainfall (mm)	Soil Fertility	Crop Yield
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

The three variables have very different numerical ranges.

- Rainfall = **280–520**
- Soil fertility = **3–9**
- Crop yield = **65–90**

Clustering Before Normalization

If clustering is performed directly on the original data, rainfall has much larger numerical values than soil fertility.

For example, a rainfall difference may be **200 mm**, while the difference in soil fertility may be only **5 points**.

Distance-based clustering algorithms such as K-Means may therefore give more importance to rainfall simply because its numerical values are larger.

As a result, the clusters may be influenced mainly by rainfall rather than by all three agricultural factors.

Clustering After Normalization

Normalization changes the variables to a common scale.

One common method is **Min-Max normalization**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

After normalization, rainfall, soil fertility and crop yield have comparable scales.

Therefore, no single variable dominates the distance calculation simply because of its larger numerical range.

Expected Clusters

After normalization, farms can be grouped based on their overall agricultural characteristics.

F1 and F4 have relatively high soil fertility and high crop yields.

F2 and F5 have high rainfall but lower soil fertility and lower crop yields.

F3 has intermediate values and can represent a middle agricultural condition.

Before vs After Normalization

Before Normalization	After Normalization
Rainfall may dominate clustering	All variables contribute more equally

Before Normalization	After Normalization
Different scales affect distance	Variables are placed on a common scale
Clusters can be biased	Clusters become more balanced
Interpretation may be less reliable	Interpretation becomes more reliable
Important features may be ignored Important features are considered together	

Importance of Normalization

Normalization is important because clustering algorithms commonly use distance calculations. If one feature has much larger numerical values than another, it can dominate the result.

After normalization, researchers can better identify relationships between **rainfall, soil fertility and crop yield**.

This can support:

- Better irrigation planning
- Soil management
- Crop selection
- Yield prediction
- Farm comparison
- Agricultural resource allocation

Conclusion

Normalization improves clustering reliability by ensuring that rainfall, soil fertility and crop yield have a more balanced influence. Therefore, the **normalized clustering result is generally more meaningful and reliable for agricultural decision-making**.

Q5. Employee Performance Evaluation

The employee dataset is:

Employee	Productivity	Attendance	Skill
E1	88	92	84

Employee	Productivity	Attendance	Skill
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

The silhouette scores provided are:

Number of Clusters	Silhouette Score
K=2	0.48

The question appears to be truncated after **K=2 = 0.48** in the latest data you pasted. In the earlier complete version, the scores were **K=2 = 0.48, K=3 = 0.68, and K=4 = 0.52**. Using that complete set:

- K=2 → 0.48
- K=3 → **0.68**
- K=4 → 0.52

Interpretation of Silhouette Score

The silhouette score measures how well employees are grouped within their clusters.

A score closer to **+1** indicates that the clusters are well separated and internally similar.

A score close to **0** indicates that clusters overlap.

A negative score can indicate poor clustering.

K = 2

The silhouette score is **0.48**.

This indicates a reasonable clustering result, but the separation between the two groups is not the best.

K = 3

The silhouette score is **0.68**, which is the **highest score**.

Therefore, K=3 provides the best clustering among the given models. It indicates that employees within each cluster are relatively similar and the clusters are reasonably well separated.

K = 4

The score decreases to **0.52**.

Although this is still acceptable, it is lower than the score for K=3. Increasing the number of clusters to four does not improve the quality of segmentation.

Most Suitable Number of Clusters

Therefore:

Optimal K = 3

The three clusters can be interpreted as:

High-performing employees

- High productivity
- High attendance
- High skill score

Average-performing employees

- Moderate productivity
- Moderate attendance
- Moderate skill level

Low-performing employees

- Lower productivity
- Lower attendance
- Lower skill score

For example, **E1 and E5** have high scores in all three measures and can be associated with the high-performing group. **E4** has comparatively low scores and can be associated with the lower-performing group. **E2 and E3** can represent the middle-performance group.

Importance for Employee Evaluation

Using three clusters helps the company understand employee performance more clearly.

High-performing employees can be recognized and rewarded.

Average-performing employees can be given motivation and development opportunities.

Low-performing employees can be provided with additional training, mentoring and performance support.

The model can therefore help management with:

- Employee recognition
- Training decisions
- Skill development
- Performance monitoring
- Resource allocation
- Career development
- Productivity improvement

Conclusion

The best model is **K=3**, because it has the highest silhouette score of **0.68**. It provides better-separated and more meaningful employee groups than K=2 or K=4. Therefore, the three-cluster solution is most suitable for effective employee performance evaluation.